

Experiment No. 3

1. Aim

Implement control structures and form validation; create a grading system based on user-entered marks.

2. Software / Tools Required

- Any text editor (VS Code / Notepad++ / Sublime Text)
- Web Browser (Google Chrome / Mozilla Firefox / Microsoft Edge)
- HTML, CSS, and JavaScript

3. Code

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Grade Calculator</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<div class="card">
  <div class="header">
    <h2>Grade Calculator</h2>
    <p class="subtitle">Enter details to calculate the grade</p>
  </div>

  <label for="name">Student Name</label>
  <input type="text" id="name" placeholder="e.g. Harry Sharma">

  <label for="prn">PRN</label>
  <input type="text" id="prn" placeholder="e.g. 12345678">

  <label for="branch">Branch / Class</label>
  <input type="text" id="branch" placeholder="e.g. Computer Engineering">

  <label for="semester">Semester</label>
  <input type="text" id="semester" placeholder="e.g. 5th Semester">

  <div class="subjects-header">Subject Marks (0-100 each)</div>

  <div class="subjects-grid">
    <div class="subject-row">
      <span class="subject-label">Maths</span>
      <input type="number" class="marks-input" id="sub1" min="0" max="100" placeholder="Marks">
    </div>
    <div class="subject-row">
      <span class="subject-label">Physics</span>
      <input type="number" class="marks-input" id="sub2" min="0" max="100" placeholder="Marks">
    </div>
    <div class="subject-row">
```

```

    <span class="subject-label">Chemistry</span>
    <input type="number" class="marks-input" id="sub3" min="0" max="100" placeholder="Marks">
  </div>
  <div class="subject-row">
    <span class="subject-label">Computer Sci</span>
    <input type="number" class="marks-input" id="sub4" min="0" max="100" placeholder="Marks">
  </div>
  <div class="subject-row">
    <span class="subject-label">English</span>
    <input type="number" class="marks-input" id="sub5" min="0" max="100" placeholder="Marks">
  </div>
</div>

<button onclick="calculateGrade()">Get Grade</button>

<div id="result"></div>
</div>

<script src="script.js"></script>
</body>
</html>

```

script.js

```

const shopName = "SneakPeak"; // const
var gst = 18; // var
let cart = []; // let

const products = [
  {
    id: 1,
    name: "Nike Air Max",
    price: 5999
  },
  {
    id: 2,
    name: "Adidas Ultraboost",
    price: 7499
  },
  {
    id: 3,
    name: "Puma RS-X",
    price: 4999
  },
  {
    id: 4,
    name: "Vans Old Skool",
    price: 4299
  }
];

function addToCart(id) {

  let product = products.find(item => item.id === id);

  let qty = Number(document.getElementById("qty" + id).value);
  let size = document.getElementById("size" + id).value;

  if (qty <= 0) {
    alert("Please enter a valid quantity.");
    return;
  }
}

```

```

// Object Destructuring
const { name, price } = product;

cart.push({
  name,
  price,
  qty,
  size
});

displayCart();

document.getElementById("qty" + id).value = 0;

alert(name + " added to cart!");
}

function displayCart() {
  let output = "";

  if (cart.length === 0) {
    document.getElementById("cartItems").innerHTML =
      "<p>Your cart is empty.</p>";

    return;
  }

  cart.forEach((item, index) => {
    output += `
    <p>
    <b>${index + 1}. ${item.name}</b><br>
    Size : ${item.size}<br>
    Quantity : ${item.qty}<br>
    Price : ₹${item.price}
    </p>
    <hr>
    `;
  });

  document.getElementById("cartItems").innerHTML = output;
}

function generateBill() {
  if (cart.length === 0) {
    alert("Your cart is empty!");

    return;
  }

  let subtotal = 0;
  let bill = `<h2>${shopName} Bill</h2>`;

  cart.forEach(item => {
    let total = item.price * item.qty;

    subtotal += total;
  });
}

```

```

        bill += `
        <p>
        ${item.name}<br>
        Size : ${item.size}<br>
        ₹${item.price} × ${item.qty} = ₹${total}
        </p>
        <hr>
        `;
    });

    let gstAmount = subtotal * gst / 100;

    let discount = 0;

    if (subtotal >= 10000) {

        discount = 1000;

    }

    let finalAmount = subtotal + gstAmount - discount;

    bill += `
    <h3>Subtotal : ₹${subtotal.toFixed(2)}</h3>

    <h3>GST (${gst}%) : ₹${gstAmount.toFixed(2)}</h3>

    <h3>Discount : ₹${discount}</h3>

    <hr>

    <h2>Total : ₹${finalAmount.toFixed(2)}</h2>

    <br>

    <h3>Thank You for Shopping!</h3>
    `;

    document.getElementById("billOutput").innerHTML = bill;
}

function clearCart() {

    cart = [];

    document.getElementById("cartItems").innerHTML =
        "<p>Your cart is empty.</p>";

    document.getElementById("billOutput").innerHTML = "";

}

```

style.css

```

* {
    box-sizing: border-box;
}

.card {

```

```

background: rgba(255, 255, 255, 0.94);
padding: 35px 45px;
border-radius: 16px;
box-shadow: 0 10px 30px rgba(0,0,0,0.35);
width: 480px;
animation: fadeIn 0.4s ease;
backdrop-filter: blur(2px);
}

body {
font-family: 'Segoe UI', Arial, sans-serif;
background: url('background.jpg');
background-size: cover;
background-position: center;
background-repeat: no-repeat;
background-attachment: fixed;
display: flex;
justify-content: center;
align-items: center;
min-height: 100vh;
margin: 0;
padding: 30px 10px;
}

@keyframes fadeIn {
from { opacity: 0; transform: translateY(10px); }
to { opacity: 1; transform: translateY(0); }
}

.header {
text-align: center;
margin-bottom: 20px;
}

h2 {
margin: 0;
color: #333;
font-size: 24px;
}

.subtitle {
margin: 6px 0 0;
color: #888;
font-size: 13px;
}

label {
display: block;
margin-top: 14px;
font-size: 13px;
font-weight: 600;
color: #444;
}

input {
width: 100%;
padding: 10px 12px;
margin-top: 6px;
border: 1.5px solid #e0e0e0;
border-radius: 8px;
font-size: 14px;
transition: border-color 0.2s, box-shadow 0.2s;
}

```

```

input:focus {
  outline: none;
  border-color: #764ba2;
  box-shadow: 0 0 3px rgba(118, 75, 162, 0.15);
}

.subjects-header {
  margin-top: 22px;
  margin-bottom: 6px;
  font-size: 13px;
  font-weight: 700;
  color: #764ba2;
  text-transform: uppercase;
  letter-spacing: 0.5px;
  border-bottom: 1.5px solid #eee;
  padding-bottom: 6px;
}

.subjects-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 12px 20px;
  margin-top: 10px;
}

.subject-row {
  display: flex;
  flex-direction: column;
  gap: 4px;
}

.subject-label {
  font-size: 13px;
  color: #444;
}

.marks-input {
  width: 100%;
  margin-top: 0;
  text-align: center;
}

button {
  width: 100%;
  margin-top: 22px;
  padding: 12px;
  background: linear-gradient(135deg, #667eea, #764ba2);
  color: #fff;
  border: none;
  border-radius: 8px;
  cursor: pointer;
  font-size: 15px;
  font-weight: 600;
  letter-spacing: 0.3px;
  transition: transform 0.15s, box-shadow 0.15s;
}

button:hover {
  transform: translateY(-1px);
  box-shadow: 0 6px 14px rgba(118, 75, 162, 0.35);
}

button:active {
  transform: translateY(0);
}

```

```

}

#result {
  margin-top: 20px;
  text-align: center;
  font-size: 14px;
  min-height: 20px;
}

.info-box {
  text-align: left;
  background: #f9fafb;
  border-radius: 8px;
  padding: 12px 14px;
  margin-bottom: 12px;
}

.info-row {
  display: flex;
  justify-content: space-between;
  padding: 3px 0;
  font-size: 13px;
  color: #444;
}

.info-row span:first-child {
  color: #888;
}

.subject-table {
  width: 100%;
  border-collapse: collapse;
  margin-bottom: 12px;
  font-size: 13px;
}

.subject-table th, .subject-table td {
  text-align: left;
  padding: 6px 8px;
  border-bottom: 1px solid #eee;
}

.subject-table th {
  color: #888;
  font-weight: 600;
  font-size: 12px;
  text-transform: uppercase;
}

.subject-table td:last-child, .subject-table th:last-child {
  text-align: right;
}

.error {
  color: #dc2626;
  font-weight: 500;
  background: #fef2f2;
  padding: 8px;
  border-radius: 6px;
  display: block;
}

.stats-row {
  display: flex;

```

```
    justify-content: space-between;
    gap: 10px;
    margin-bottom: 12px;
}

.stat-box {
    flex: 1;
    background: #f0f4ff;
    border-radius: 8px;
    padding: 8px;
}

.stat-label {
    font-size: 11px;
    color: #888;
    text-transform: uppercase;
}

.stat-value {
    font-size: 16px;
    font-weight: 700;
    color: #444;
    margin-top: 2px;
}

.success-box {
    padding: 14px;
    border-radius: 8px;
    background: #f0fdf4;
    border: 1px solid #bbf7d0;
}

.grade-badge {
    display: inline-block;
    margin-top: 4px;
    padding: 6px 18px;
    border-radius: 20px;
    background: linear-gradient(135deg, #16a34a, #22c55e);
    color: #fff;
    font-size: 18px;
    font-weight: 700;
    letter-spacing: 1px;
}

.grade-badge.fail {
    background: linear-gradient(135deg, #dc2626, #ef4444);
}
```

4. Output



Student Grading System

Manage & Evaluate Student Performance with a fresh, modern experience

SMART GRADING WORKFLOW

Grade students faster with a polished report experience

Enter student details, add marks, and generate a clear summary instantly.

Tanshi Dodiya • PIN 24070521021

5 subjects

Instant results

Validated inputs

STUDENT FORM

Fill in the required information below to create a polished student report card.

Student Details

Mahek Malkan
24070521040

Student Information

Student Name *

Mahek Jatin Malkan

Roll Number *

24070521040

Class *

10th

Section *

A

Email Address *

mahek.malkan.batch2024@sinagpur.siu.edu.in

Phone Number *

8788419284

Marks (Out of 100)

English *

89

Mathematics *

45

Science *

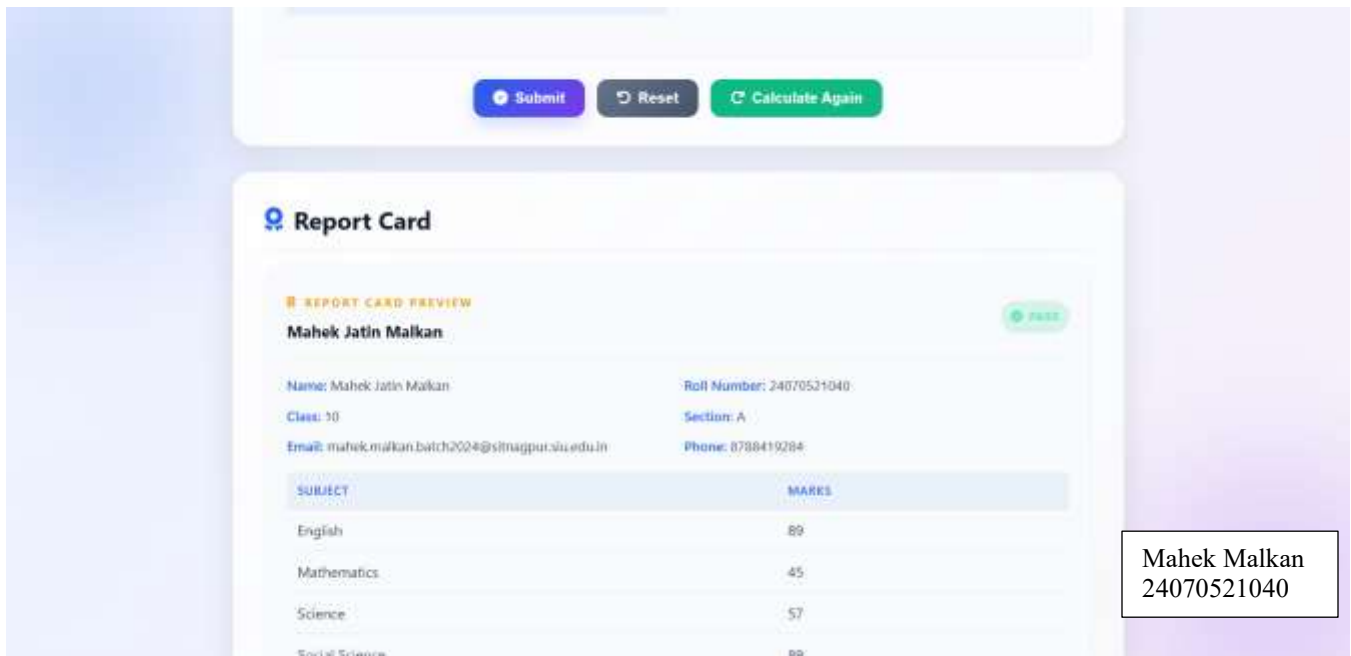
67

Social Science *

89

Computer *

Mahek Malkan
24070521040



Mahek Malkan
24070521040



Mahek Malkan
24070521040

5. Case Study

index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      Student Grading System
    </title>

    <body>
      <h2>Student Grading System</h2>
      <form>
```

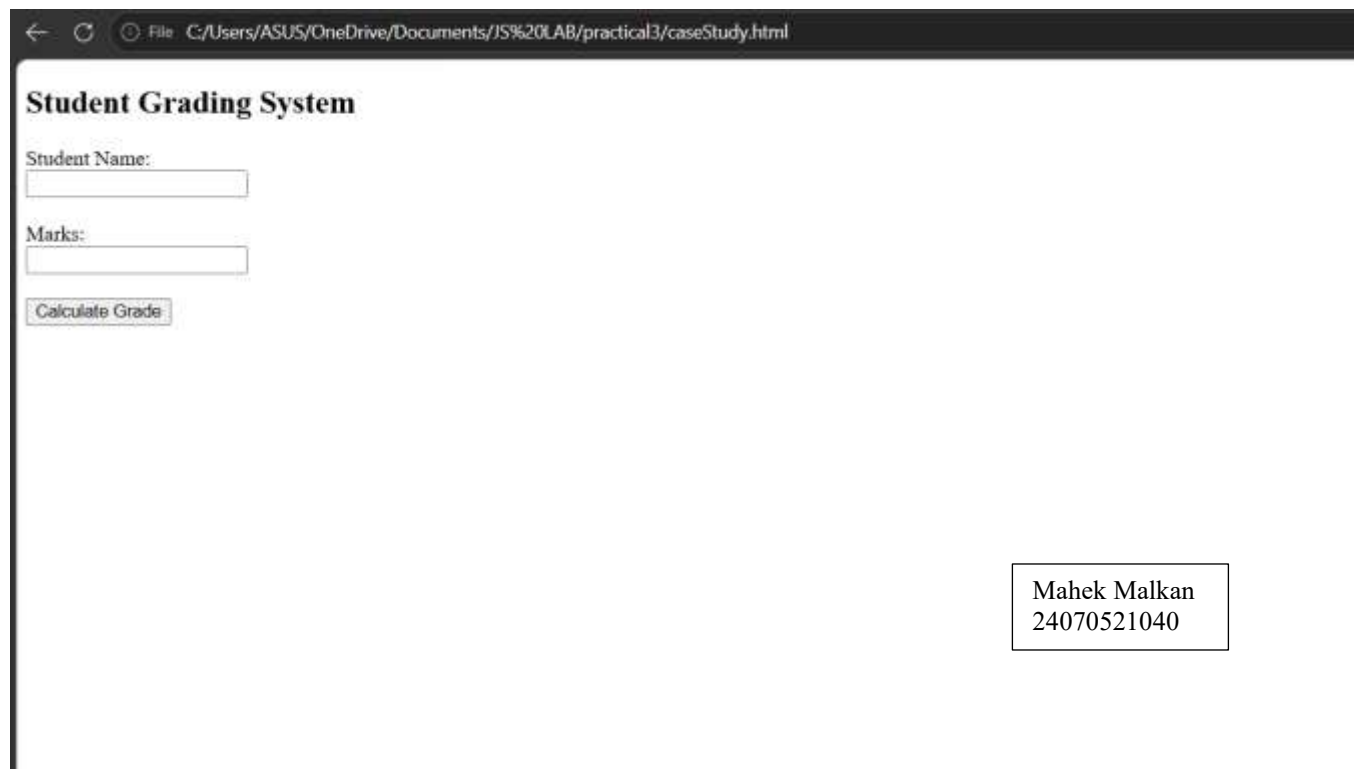
```
<label>Student Name: </label><br>
<input type="text" id="name"><br><br>

<label>Marks:</label><br>
<input type="number" id="marks"><br><br>

<input type="button" value="Calculate Grade" onclick="gradeSystem()">
</form>
</body>

</head>
</html>
```

5. Case Study Output



The screenshot shows a web browser window with the address bar displaying the file path: C:/Users/ASUS/OneDrive/Documents/JS%20LAB/practical3/caseStudy.html. The page title is "Student Grading System". The form contains two input fields: "Student Name:" and "Marks:". Below these fields is a button labeled "Calculate Grade".

Student Name:

Marks:

Calculate Grade

Mahek Malkan
24070521040